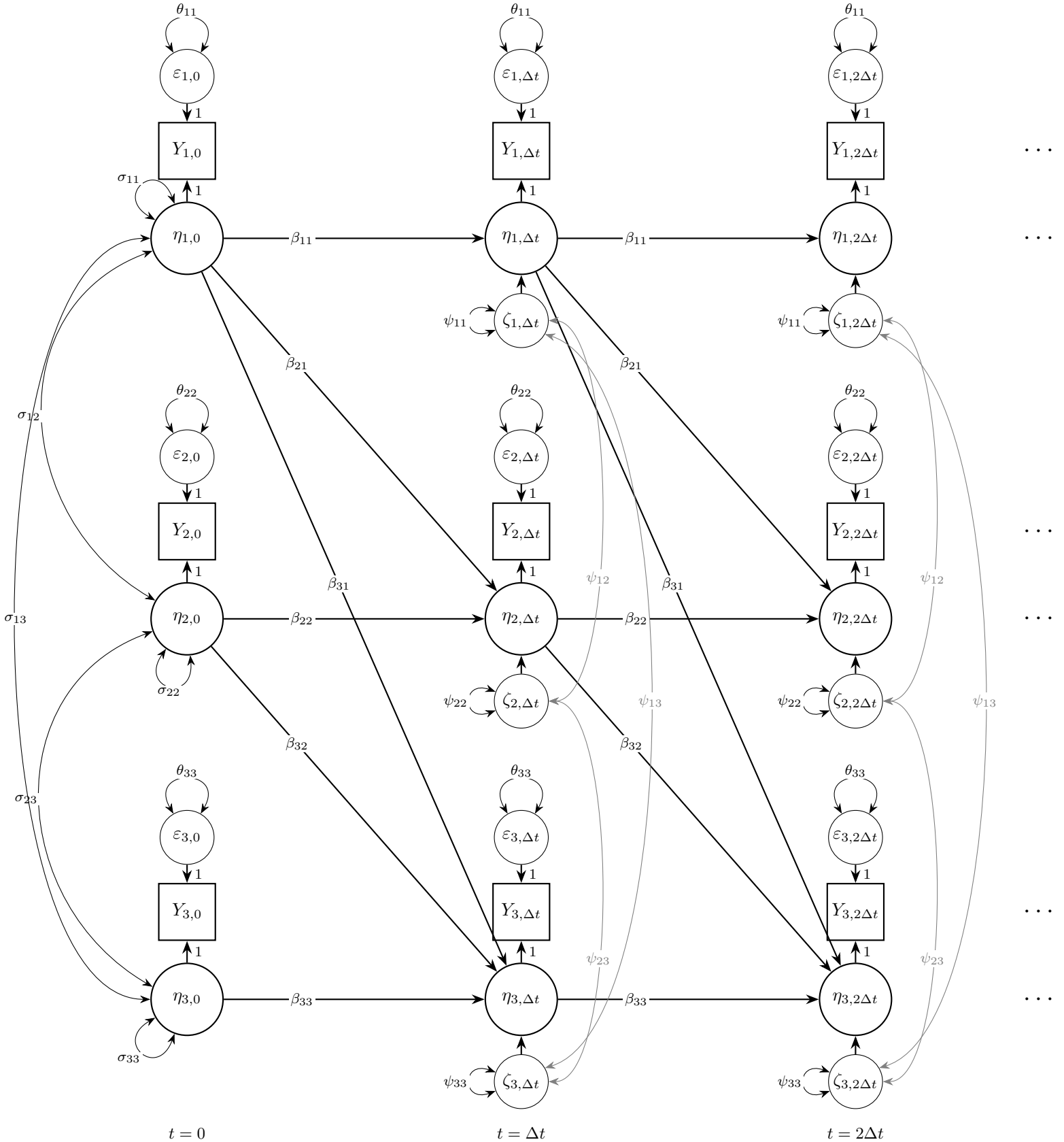



$\boldsymbol{\eta}_t = (\eta_{1,t}, \eta_{2,t}, \eta_{3,t})'$ (latent), $\quad \mathbf{y}_t = (Y_{1,t}, Y_{2,t}, Y_{3,t})'$ (observed), $\quad$ occasions $\Delta t$ apart $(t = 0, \Delta t, 2\Delta t, \ldots)$

$$\boldsymbol{\Phi} = \begin{pmatrix} \phi_{11} & 0 & 0 \\ \phi_{21} & \phi_{22} & 0 \\ \phi_{31} & \phi_{32} & \phi_{33} \end{pmatrix} \qquad \boldsymbol{\Sigma}\text{: the covariance matrix of the Wiener process per unit time}$$

$$\boldsymbol{\beta} = \exp(\boldsymbol{\Phi}\Delta t) = \begin{pmatrix} \beta_{11} & 0 & 0 \\ \beta_{21} & \beta_{22} & 0 \\ \beta_{31} & \beta_{32} & \beta_{33} \end{pmatrix} \qquad \boldsymbol{\Psi} = \int_0^{\Delta t} \exp(\boldsymbol{\Phi} s)\, \boldsymbol{\Sigma} \exp(\boldsymbol{\Phi}' s)\, \mathrm{d}s = \begin{pmatrix} \psi_{11} & \psi_{12} & \psi_{13} \\ \psi_{21} & \psi_{22} & \psi_{23} \\ \psi_{31} & \psi_{32} & \psi_{33} \end{pmatrix}$$

$$\boldsymbol{\Lambda} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad \boldsymbol{\Theta} = \begin{pmatrix} \theta_{11} & 0 & 0 \\ 0 & \theta_{22} & 0 \\ 0 & 0 & \theta_{33} \end{pmatrix}$$

$$\boldsymbol{\eta}_0 \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0), \quad \boldsymbol{\mu}_0 = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{pmatrix}, \quad \boldsymbol{\Sigma}_0 = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{pmatrix}$$